Intel Penryn
Intel's Penryn quad-core features a whopping 12 MB L2 cache


Intel's X58 platform is the first to officially support both SLI and Crossfire

The initial processors came with DDR2 memory controllers and were designed for AM2 and AM2+ sockets. This is another plus for AMD – no backward compatibility issues with newer processors and older motherboards. Newer Phenom 2 processors for Socket AM3 motherboards have also made it to market and these processors have an upgraded memory controller that supports DDR2 as well as DDR3. And the beauty of this is that Socket AM3-based Phenom 2 processors can be used in AM2+ as well as AM3 boards simply because the CPU sockets are pin-compatible. Of course there is a slight difference in the AM3 processor's L3 cache and memory controller (these parts are collectively referred to as the *uncore*). They run at a speed of 2.0 GHz. The non-AM3 Phenom 2s run at a speed of 1.8 GHz.

The DDR3 memory controller supports up to 1333 MHz memory clock speeds. On an AM3 motherboard, the memory bandwidth is higher (21 GB/s as opposed to 17.1 GB/s). Since the AM3 socket also supports DDR2 memory speeds of up to 1066 MHz, motherboard manufacturers have an interesting choice of whether to provide a motherboard with DDR2 or DDR3 memory slots, or perhaps both, and this is where the platform's true flexibility lies. This is an attractive plus especially when compared to the competition. Intel has an LGA 775 interface for all its Core 2 Duos and Quads but the new Core i7 CPUs need investment in a completely new (and expensive) LGA 1366 motherboard, not to mention a mandatory investment in DDR3 memory – more arm-twisting due to near monopolistic market conditions?

In terms of clock speeds, the fact that AMD was able to scale its quad-cores up to 3.2 GHz from a previous high of 2.6 GHz was of interest to us; as was the jump from 65nm to 45nm since this brings its manufacturing process on a par with the competition. Earlier we felt that a meagre 2 MB of L3 cache was one of the chinks in Phenom's armour. The 6 MB cache on the higher end Phenom 2s would surely yield favourable results. Then there was the sudden tilt to DDR3. DDR3 offers greater bandwidth in comparison to DDR2. We're seeing faster, lower latency DDR3 chips slowly emerge and these CPUs could scale to higher performance levels with faster memory; of course, so would Intel's Nehalem.

BARCELONA BRIEFS

For AMD, the Barcelona architecture (known as K10 and the processors are called Phenom), represented a big improvement from its Athlon 64 (K8) architecture and is the basis for the Phenom 2 CPUs we're seeing today. Besides being a native quad-core (Athlon 64s were native dual-cores) Phenom had the ability to process 128-bit SSE operations without the need for breaking them up into two 64-bit operations as with the Athlon 64s. This also frees up a decode port since one operation results in one micro-op, whereas two 64-bit operations would result in two. The extra decode bandwidth available meant that AMD had to tweak various other parts of the core to prevent bottlenecks. Keeping wider instruction units fed means having more instructions in flight and this makes mandatory a better branch prediction unit — to predict branches that have labels (direct branching) as well as branches that point to a memory location (indirect branches). The ability to track more branches equates to greater accuracy since there's more historic data present for reference.

A major plus for the Core architecture was its ability to allow *load* instructions to bypass other *load*s and instructions already in the pipe. Since one third of the instructions in a program are typically *load*s, better *load*s performance means better application performance. One breakthrough was the ability to push *load*s ahead of stores. This was previously impossible due to the fact that an earlier store could cause a data mismatch with the current *load*. The hit taken in such a data mismatch is actually quite small (Intel puts it between one and two percent). This reordering of loads is something Barcelona copied from Core 2. Barcelona can also execute *load*s ahead of store if the two don't share the same memory address (thereby bringing chances of a data mismatch). Barcelona CPUs also feature an improved memory controller over Athlon 64's controller and instead of a single 128-bit controller there are two 64-bit controllers making things a little more efficient. Obviously DDR2 and DDR3 support were also built into the controllers for future compatibility. One of the most noticeable additions was 2 MB of L3 cache that was common to all four cores and this is a necessity when having multi-core CPUs — after all there has to be a common pool of memory for each of the cores to work together more efficiently.

While Barcelona itself outperformed the Athlon 64 CPUs it did not come close to Intel's dual- and quad-core CPUs based on the Core architecture. But as with everything historic — it paved the way for something better.

CPU SPECIFICATIONS

Processor	AMD Phenom 2	AMD Phenom	Intel Core i7	Intel Core 2 Quad (Penryn)
Manufacturing Process	45nm	65nm	45nm	45nm
L1 Cache	64K + 64K per core	64K + 64K per core	32KB + 32KB per core	32KB + 32KB per core
L2 Cache	512KB per core	512KB per core	256KB per core	2 x 6 MB
L3 Cache	6MB	2MB	8MB	-
Transistor Count (in millions)	758	450	731	820
Die Size	258 mm2	285 mm^2	263 mm^2	214 mm^2
Max Clock Speeds	3.2 GHz	2.6 GHz	3.2 GHz	3.2 GHz
Thermal Design Power (TDP)	125 / 95 watts	125 / 95 watts	130 watts	130 / 95 watts